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BIOSENSIA-DC: A DRUGCLIP-BASED COMPUTATIONAL PROTOTYPE FOR TARGET FISHING APPLIED TO BIORECEPTOR DISCOVERY

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Abstract.
To be written after the main text. Suggested content:
- Introduce BioSensIA as a project aimed at supporting the discovery of biorecognition elements for biosensors.
- Present BioSensIA-DC as a computational prototype for target fishing: given a query molecule, rank candidate protein pockets.
- Explain that the prototype adapts DrugCLIP, originally designed for virtual screening, to the reverse direction: molecule-to-pocket retrieval.
- Highlight implementation contributions: construction of candidate-pocket LMDB files, query-molecule LMDB files, molecule lookup via SMILES/PubChem, target-fishing retrieval, and an initial benchmarking workflow.
- State clearly that the current quantitative results are preliminary and unsatisfactory, including for the original DrugCLIP model in the local evaluation setup.
- Frame the paper as a methodological and computational-prototype report, not as a validated performance claim.

Keywords: *Biosensors, Target fishing, DrugCLIP, Contrastive learning, Chemoinformatics*

1. INTRODUCTION

1.1 Biosensors and the bioreceptor-selection problem

- Explain that biosensors rely on a biorecognition element capable of interacting selectively with the analyte of interest. Examples include enzymes, antibodies, aptamers, receptors, and other biomolecular recognition structures.
- Emphasize that selecting a suitable bioreceptor is a critical step in biosensor development. This choice is especially difficult for small molecules, for analytes with scarce experimental data, and for sensing problems where the relevant biological interactions are not obvious.

- Present the computational problem addressed by BioSensIA: given an analyte molecule, identify proteins, protein domains, binding pockets, or biomolecular structures that may interact with it and therefore deserve investigation as candidate bioreceptors.
- Explain that BioSensIA-DC is one computational component of this broader project. Its role is to rank candidate protein pockets for a query molecule using learned molecular and pocket representations.

1.2 Motivating example: okadaic acid

- Introduce okadaic acid as a motivating analyte because of its relevance to diarrhetic shellfish poisoning and marine-toxin monitoring.
- Explain that okadaic acid is known to interact with serine/threonine phosphatases, especially PP2A, and that structural data exist for PP2A–okadaic-acid complexes.
- Use this example to motivate target fishing in the biosensor context: if a molecule is known or suspected to bind a biomolecular structure, a computational system that ranks possible targets may help guide the search for biorecognition elements.
- Make clear that the prototype is not limited to okadaic acid. The intended input is any query molecule for which a user wants to explore plausible protein-pocket candidates.

1.3 Virtual screening, target fishing, and representation learning

- Define virtual screening as the task in which a target, usually a protein pocket, is used as a query to rank candidate molecules.
- Define target fishing as the reverse task: a molecule is used as a query, and candidate targets, proteins, or pockets are ranked according to their predicted compatibility with that molecule.
- Briefly mention established target-fishing approaches such as reverse pharmacophore mapping and ligand-similarity-based prediction. These methods provide conceptual baselines for the task, even though BioSensIA-DC follows a different representation-learning approach.
- Introduce DrugCLIP as a contrastive model that learns compatible representations of protein pockets and molecules. DrugCLIP formulates virtual screening as dense retrieval.
- Introduce Uni-Mol as the 3D molecular representation framework used by DrugCLIP for molecules and pockets.
- Explain the central idea behind BioSensIA-DC: because DrugCLIP learns a similarity score between molecules and pockets, the same score can be used in the reverse direction for target fishing.

1.4 Objectives and contributions

- State the main objective of the paper: to describe BioSensIA-DC, an open computational prototype that extends DrugCLIP for molecule-to-pocket target fishing in the context of bioreceptor discovery.
- List the implemented contributions:
  - construction of a candidate-pocket LMDB file from DrugCLIP-compatible data;
  - construction of query-molecule LMDB files from SMILES strings, local DrugCLIP identifiers, PubChem CIDs, or compound names;
  - implementation of molecule-to-pocket retrieval;
  - generation of ranked pocket lists with compatibility scores;
  - implementation of an initial benchmarking and diagnostic workflow;
  - documentation of installation and execution steps in a public repository.
- State the main limitation upfront: suitable benchmark validation has not yet been completed, and the preliminary benchmark results are poor both for the fine-tuned BioSensIA-DC model and for the original DrugCLIP model under the current local evaluation setup.
- Explain how the paper should therefore be interpreted: as a computational-prototype and methodology paper, not as evidence that the current model is already a reliable target-fishing predictor.

2. METHODOLOGY

2.1 Overview of the BioSensIA-DC workflow

- Describe BioSensIA-DC as a prototype derived from DrugCLIP, with additional code for the target-fishing direction.
- Present the general workflow:
  1. receive one or more query molecules;
  2. build or load a DrugCLIP-compatible molecule LMDB file;
  3. build or load a candidate-pocket LMDB file;
  4. encode molecules and pockets using the DrugCLIP encoders;
  5. compute molecule-pocket similarity scores;
  6. rank candidate pockets by score;
  7. save the ranked list for downstream analysis.

- Include a schematic figure showing the molecule-to-pocket direction:
  - query molecule on the left;
  - molecule encoder;
  - cached candidate-pocket embeddings;
  - similarity computation;
  - ranked list of protein pockets on the right.
- Mention the public repository:
  - `https://github.com/fnaufel/BioSensIA-DC`
- Explain that the repository contains source code, installation instructions, retrieval scripts, target-fishing scripts, and an initial benchmarking workflow.

2.2 Base model: DrugCLIP and Uni-Mol

- Describe DrugCLIP as a two-tower architecture:
  - one encoder maps molecules to latent vectors;
  - the other encoder maps protein pockets to latent vectors;
  - a similarity score compares the two vectors.
- Explain that both encoders are based on Uni-Mol-style 3D representations, using atom types and atomic coordinates.
- Define the conceptual similarity score:

$$s(p, m) = g_\phi(x^p)^T f_\theta(x^m) \tag{1}$$

- Explain the notation:
  - p denotes a protein pocket;
  - m denotes a molecule;
  - x^p and x^m denote their input representations;
  - g_ϕ is the pocket encoder;
  - f_θ is the molecule encoder;
  - $s(p, m)$ is a compatibility score, not a calibrated binding affinity.
- Explain that virtual screening ranks molecules for a fixed pocket:

$$p \to \{m_1, m_2, \ldots, m_n\} \tag{2}$$

- Explain that target fishing reverses the retrieval direction:

$$m \to \{p_1, p_2, \ldots, p_n\} \tag{3}$$

- Emphasize that BioSensIA-DC preserves the DrugCLIP model architecture and changes the surrounding data preparation, retrieval, and fine-tuning workflow.

2.3 Data organization and LMDB files

- Explain that DrugCLIP stores molecules and pockets in LMDB files, with atom types, coordinates, and auxiliary identifiers.
- Describe the original DrugCLIP data organization:
  - molecule libraries for virtual screening;
  - pocket data extracted from protein-ligand complexes;
  - model checkpoints;
  - atom dictionaries;
  - cached embeddings.
- Explain why target fishing requires a different organization: pockets become the candidate library, while molecules become the query objects.
- Describe the candidate-pocket LMDB construction:
  - read DrugCLIP-compatible structural records;
  - extract pocket entries;
  - preserve PDB identifiers;
  - write a new LMDB containing candidate pockets;
  - allow later embedding precomputation and caching.
- Describe the query-molecule LMDB construction:
  - accept SMILES strings directly;
  - accept local DrugCLIP molecule identifiers;
  - accept PubChem CIDs;
  - accept PubChem compound names;
  - convert the query into a DrugCLIP-compatible LMDB record.
- Discuss molecular-standardization issues that must be handled carefully:
  - salts versus parent compounds;
  - tautomeric forms;
  - stereochemistry;
  - conformer generation;
  - consistency between training, query, and benchmark data.

2.4 Target-fishing retrieval implementation

- Identify the main implemented modules:
  - `biosensia_target_fishing.py`;
  - `biosensia_retrieval.py`;
  - `lmdb_helpers.py`;
  - `biosensia_target_fishing_benchmark.py`.
- Describe the retrieval procedure:
  1. load query molecules from an LMDB file;
  2. load candidate pockets from an LMDB file;
  3. encode query molecules;
  4. encode candidate pockets or reuse cached pocket embeddings;
  5. compute the similarity matrix;
  6. rank pockets by compatibility with the query molecule;
  7. write the ranked output to a tabular file.
- Explain how cached embeddings reduce repeated computation. When the same candidate-pocket library is reused, only the query molecules need to be newly encoded.
- Explain that the output score should be interpreted as a ranking score. It is not a binding free energy, an inhibition constant, or a direct physical measurement.
- Include a planned example command or code fragment:

```
import biosensia_target_fishing as tf

tf.create_mol_lmdb("okadaic acid", "data/oka.lmdb")
tf.build_candidate_pockets_lmdb("data/candidate_pockets.lmdb")
```

- Include another planned example for running retrieval:

```
cd external/DrugCLIP
./target_fishing.sh
```

2.5 Fine-tuning formulation for target fishing

- Explain that the original DrugCLIP contrastive loss is naturally aligned with virtual screening, but not necessarily optimized for target fishing.
- Describe the issue with repeated ligands:
  - the same molecule can appear in several positive protein-ligand complexes;
  - those complexes may correspond to different structures, homologous proteins, isoforms, or related binding sites;
  - in target fishing, all known pockets for the same molecule should be treated as positives;
  - the original diagonal-only contrastive loss may treat some known positives as implicit negatives, depending on batch composition.
- Define the set of known positive pockets for a molecule:

$$P^{+}(m) = \{p : (p, m) \in D\} \tag{4}$$

- Explain that D denotes the available set of observed positive pocket-molecule pairs.
- Define a positive-pair mask for a batch:

$$Y_{ij} = \begin{cases} 1, & \text{if } p_j \in P^{+}(m_i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

- Define the multi-positive molecule-to-pocket loss:

$$L_i^{m \to p} = -\log \frac{\sum_j Y_{ij} \exp(S_{ij}/\tau)}{\sum_j \exp(S_{ij}/\tau)} \tag{6}$$

- Explain that $S_{ij} = s(p_j, m_i)$ is the similarity between pocket p_j and molecule m_i, and that τ is the contrastive temperature.
- Explain the intuition: for each query molecule, the loss increases the total softmax probability assigned to all known positive pockets in the batch.
- Present fine-tuning variants to be discussed:
  - freeze both encoders and train only projection layers;
  - train projection layers plus the temperature parameter;
  - unfreeze upper encoder layers;
  - train the full model with a smaller learning rate;
  - compare molecule-to-pocket loss alone with a symmetric loss that also includes pocket-to-molecule terms.
- State that the current implementation and experiments are preliminary and that the poor validation results do not yet support a performance claim.

2.6 Benchmarking and diagnostic evaluation

- Explain that benchmarking target fishing is difficult because a query molecule may have several true targets, and missing pairs are not necessarily true negatives.
- Describe the implemented benchmark workflow:
  - construct an explicit table of positive molecule-pocket pairs;
  - build query-molecule LMDB files;
  - build candidate-pocket LMDB files;
  - run molecule-to-pocket retrieval;
  - compare ranked lists against known positives;
  - compute ranking metrics.
- List metrics to be discussed:
  - top-k accuracy;
  - recall@k;
  - mean reciprocal rank;
  - rank of the first known positive;
  - enrichment-style metrics for early retrieval;
  - AUROC only with caution, because it may be less informative when early ranking is the main objective.
- Explain that preliminary results are poor for both the fine-tuned BioSensIA-DC model and the original DrugCLIP model in the current evaluation setup.
- Explain that this suggests the need for diagnostic work before any strong conclusion:
  - verify data splits;
  - verify checkpoint compatibility;
  - verify LMDB contents;
  - verify atom dictionaries;
  - verify cached embeddings;
  - verify whether candidate pockets match the positive-pair table;
  - verify whether the benchmark task is consistent with the model's training assumptions.

2.7 Computational environment and reproducibility

- Describe the computational environment used for development:
  - GNU/Linux;
  - Python 3.11;
  - PyTorch with CUDA;
  - RDKit;
  - Uni-Core;
  - DrugCLIP;
  - LMDB.
- Explain the use of `uv` for Python environment management.
- Explain that a GPU is required by inherited constraints in the original DrugCLIP code path.
- Discuss technical issues encountered:
  - compiling Uni-Core;
  - CUDA extension compatibility;
  - C++ standard-library compatibility;
  - GPU memory limits;
  - long-running embedding precomputation;
  - embedding-cache invalidation.
- Explain why these details matter: they affect reproducibility and practical adoption of 3D deep-learning models in computational biosensor research.

3. RESULTS AND DISCUSSION

3.1 Implemented software artifacts

- Present the public repository as the main concrete result of this stage.
- List implemented artifacts:
  - query-molecule LMDB generation;
  - candidate-pocket LMDB generation;

  - molecule-to-pocket retrieval;
  - ranked-pocket output;
  - preliminary benchmark code;
  - installation and execution documentation.

- Include a table with columns such as:
  - file or module name;
  - purpose;
  - input;
  - output;
  - current status.

- Discuss how these artifacts convert DrugCLIP from a virtual-screening-only workflow into a prototype that can also run target-fishing queries.

3.2 Qualitative query example with okadaic acid

- Present okadaic acid as a qualitative sanity-check query.
- Explain the biological expectation: PP2A-related pockets should be considered relevant because PP2A is a known okadaic-acid target.
- Include a planned table with:
  - rank;
  - PDB identifier;
  - score;
  - protein or pocket annotation;
  - whether the result is known, plausible, unclear, or likely irrelevant.

- Discuss whether PP2A-associated structures appear near the top of the ranking.
- State clearly that this qualitative example is not a benchmark. It is only a sanity check and an illustration of the workflow.

3.3 Preliminary quantitative results

- Report that the quantitative results obtained so far are unsatisfactory.
- Separate two observations:
  - the fine-tuned BioSensIA-DC model currently performs poorly;
  - the original DrugCLIP model also performs poorly under the current local benchmark workflow.
- Avoid claiming improvement over DrugCLIP.
- Present the results as a diagnostic table, not as a final performance table. Suggested columns:
  - model;
  - checkpoint;
  - query set;
  - candidate-pocket set;
  - metric values;
  - diagnostic comment.
- Explain that the current evidence points to unresolved issues in reproduction, data handling, benchmark construction, or model adaptation.

3.4 Possible causes of the poor benchmark results

- Discuss possible data problems:
  - mismatch between training, validation, and candidate LMDB files;
  - differences between the downloaded data and the data used in the original DrugCLIP paper;
  - duplicated PDB identifiers with different structures;
  - ligand-identity inconsistencies;
  - salt, tautomer, stereochemistry, or conformer inconsistencies.
- Discuss possible implementation or protocol problems:
  - wrong or incompatible checkpoint;
  - stale cached embeddings;

  - incorrect correspondence between pocket identifiers and benchmark positives;
  - candidate libraries that do not contain the expected positives;
  - metrics computed over an inappropriate candidate set;
  - target-fishing evaluation derived from data originally organized for virtual screening.

- Discuss possible modeling limitations:
  - DrugCLIP was primarily motivated by virtual screening;
  - target fishing is multi-positive by nature;
  - unobserved pairs are not reliable true negatives;
  - the similarity score is not physically calibrated;
  - fine-tuning projection layers alone may be insufficient.

- Conclude that the next step must be a systematic audit before interpreting the model's predictive value.

3.5 Relationship to existing target-fishing tools

- Compare BioSensIA-DC conceptually with PharmMapper:
  - PharmMapper performs reverse pharmacophore mapping;
  - BioSensIA-DC performs learned molecule-pocket representation matching;
  - both produce ranked target-candidate lists.

- Compare BioSensIA-DC conceptually with SwissTargetPrediction:
  - SwissTargetPrediction relies on similarity to known ligands;
  - BioSensIA-DC relies on learned 3D compatibility between molecules and pockets;
  - the two approaches may be complementary.

- Discuss how future versions of BioSensIA could combine rankings from multiple methods:
  - rank aggregation;
  - consensus among tools;
  - grouping by protein family;
  - docking-based follow-up;
  - literature-based validation.

3.6 Relevance to the BioSensIA project

- Explain that BioSensIA-DC is one module in a larger computational pipeline for biosensor-oriented bioreceptor discovery.
- Relate the prototype to planned BioSensIA components:
  - UniProt and Gene Ontology queries;
  - domain and family analysis;
  - PDB and AlphaFold structural data;
  - pocket prediction;
  - docking and pharmacophore analysis;
  - evidence aggregation for candidate bioreceptors.
- Explain that even before full validation, BioSensIA-DC provides an operational way to turn a molecule-centered question into a ranked list of structural candidates.
- Emphasize that the current results motivate further validation rather than immediate deployment.

4. CONCLUSIONS

4.1 Summary of contributions

- Summarize that the paper presented BioSensIA-DC, a DrugCLIP-based computational prototype for molecule-to-pocket target fishing.
- Highlight the main engineering contribution: implementing the reverse retrieval direction needed to rank candidate pockets for a query molecule.
- Highlight that the prototype includes data preparation, retrieval execution, ranked output generation, and preliminary evaluation tools.
- Highlight that the implementation is publicly available and intended to support reproducible development.

4.2 Limitations

- State that the current model has not yet been validated as a reliable predictor.
- State that preliminary benchmark results are poor for both the fine-tuned model and the original DrugCLIP model in the current local evaluation setup.
- Explain that these results may reflect unresolved problems in reproduction, data consistency, benchmark construction, or the adaptation of DrugCLIP to target fishing.
- State that no claim of improved predictive performance is made in the present paper.

4.3 Future work

- Reproduce the original DrugCLIP benchmarks more carefully before interpreting target-fishing results.
- Audit checkpoints, LMDB files, atom dictionaries, cached embeddings, and benchmark-positive definitions.
- Build a target-fishing benchmark with rigorous train-test separation and clear ligand-identity policies.
- Evaluate multi-conformer query strategies, including maximum, mean, median, quantile, and log-sum-exp aggregation.
- Expand the candidate-pocket library using PDB, BioLiP, sc-PDB, AlphaFold structures, and pocket-prediction tools.
- Compare BioSensIA-DC rankings with PharmMapper, SwissTargetPrediction, and docking-based validation.
- Integrate BioSensIA-DC into a broader BioSensIA interface for biosensor-oriented candidate discovery.

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- Include the required generative-AI disclosure. Suggested wording:
  - The author used ChatGPT (OpenAI) to assist with manuscript organization, language revision, and discussion of alternative formulations. All scientific content, interpretation of results, and conclusions remain the responsibility of the author.

REFERENCES